



Priniples of Balanced Ligamentous Tension (BLT)

OMM

Hugh Ettlinger, DO, FAAO
Associate Professor, OMM

Session Objectives

- Understand the Anatomic and physiologic basis of BLT
- Understand the concept of a Ligamentous Articular Mechanism
- Understand the role of sensing in the performance of Osteopathic techniques
- Understand the “natural agencies” responsible for correction in BLT
- Understand the use of indirect and direct approaches in BLT



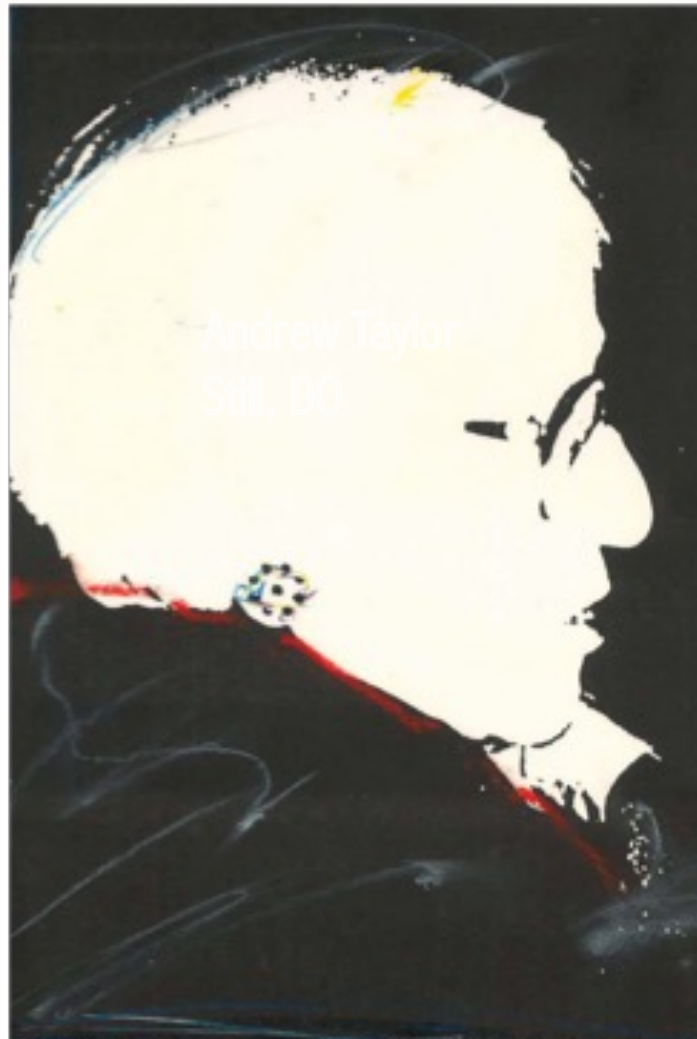
BLT is the treatment principle taught by W.G. Sutherland.

Dr. Sutherland graduated from the ASO in 1900, and credits Dr. Still for his approach



“Dr. Still has taken my hand in his and allowed me to feel the lesion as it was being exaggerated and then as the natural agencies pulled the bones back into place.”

COT p 112





Anne Wales, DO,
who was Dr.
Sutherland's
student and part
of his teaching
faculty for 10
years.

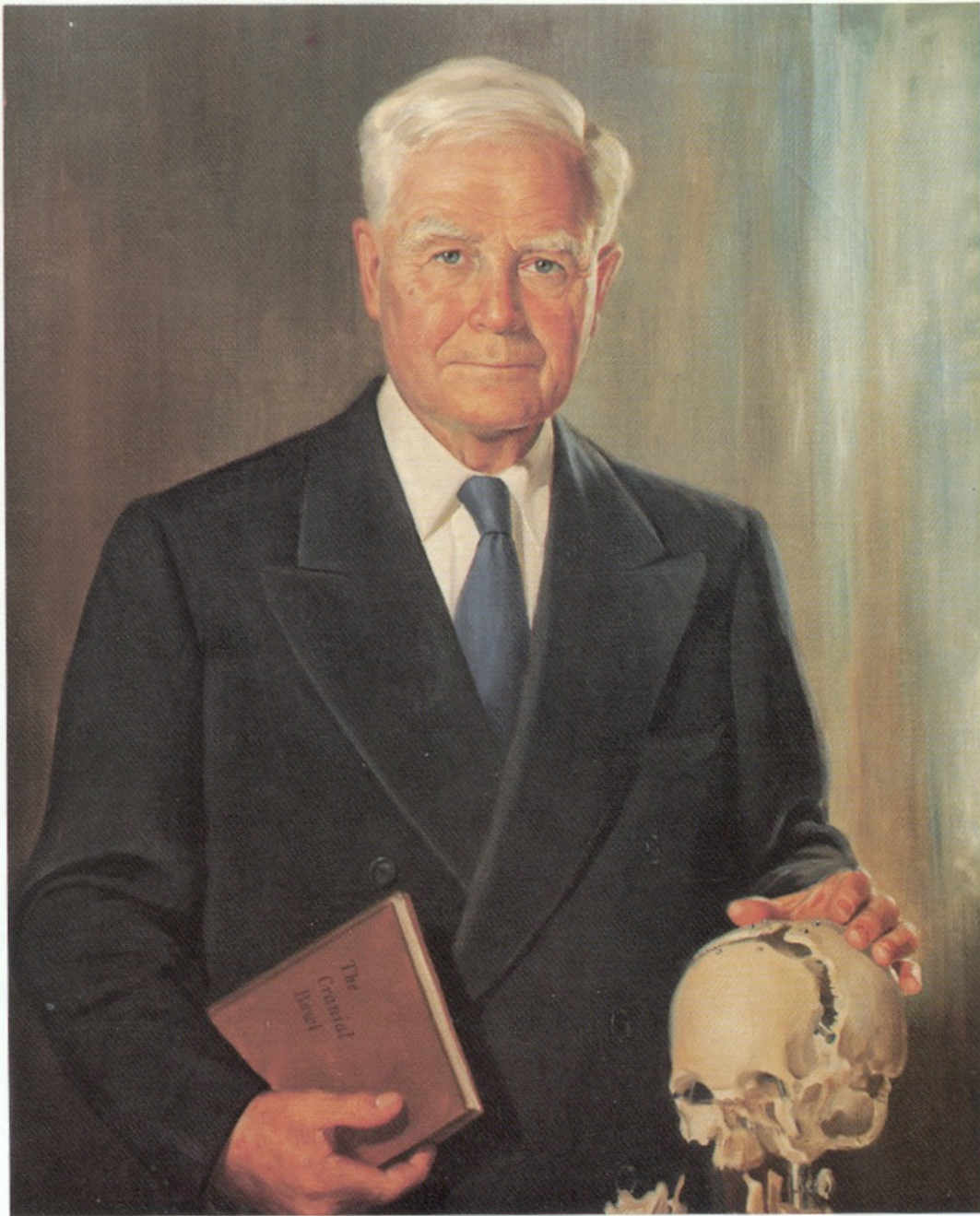
She spent 18
years teaching us
in the study
group.

Introduction

- Studying Osteopathy is more than learning techniques
 - Philosophy/principles
 - Anatomy/Physiology
 - Medicine

Introduction - Skills

- OMM is a diagnostic art and science where you use your Osteopathic skills to understand your patient's problem more clearly
- It is about what you **feel**, not what you **do**!



“If you understand
the mechanism, the
treatment is simple”

W. G. Sutherland

William Garner Sutherland, D.O., D.Sc.(hon.)

- I could teach a chimpanzee to treat!
- The job of the physician is to make the diagnosis.
 - STAN SCHIOWITZ, DO

Introduction

- Thinking vs. Tinkering...
- Learning Technique
- Doing vs. sensing
- Learn to continue sensing as you shift to treatment.
- Observe the potency within the patient's body that is the motive power for all therapeutic changes

- Diagnosis and treatment are inseparable. This does not mean you begin treating without a diagnosis.
- Remain in a perceptive state during the treatment, continue to look for diagnostic clues in the response of the patient to the treatment.
- Place your attention on the patient's response to your treatment, not on what you are doing.

Powers Within the Patient

- “A Physician who practices Osteopathy must administer the treatment with his own manual operations or he will not be aware of the information that the process discloses”
 - A.L.Wales, DO

Conclusions

- “As the treatment program progresses his experience as operator and the continuing diagnostic interpretation as his guide inspire this creation.”
– A.L. Wales, DO



Introduction

- The better you understand the problem, as well as the patient who has the problem, the more effective your treatment will be.
- The treatment plan is a critical step
- You strive to understand the subtle differences in each patient's presentation to create an individualized treatment plan, there are no algorithms or protocols!!
- Clear mental picture of anatomy



No two patients with the same disease present the same way. No patient with the same disease presents the same way on different visits.

A.L. Wales, DO

Ligamentous Articular Mechanism

- “In describing spinal lesions, I prefer the term ‘ligamentous articular strains’; for lesions in the cranium, ‘membranous articular strains’. The spinal lesion includes the ligaments as well as the joints. The cranial lesion includes the intracranial membranes as well as the articulations. ”

W.G.Sutherland, DO
TSO

Ligamentous Articular Mechanism

- BLT vs. LAS

Ligamentous Articular Mechanism

- “The ligaments in their tension regulate the movement in the spinal articulations, as check agents to voluntary muscular action. They might be called ‘reciprocal tension ligaments’.

W.G.Sutherland, DO

TSO

Ligamentous-Articular Mechanism

- “The neuromuscular system provides the power and control for voluntary mobility of our skeletal system. Underlying this, however, is an involuntary action within the skeletal system itself. Mechanical problems in this action are the first consideration of a working diagnosis for an Osteopath.”
- A.L. Wales, DO SML 1993

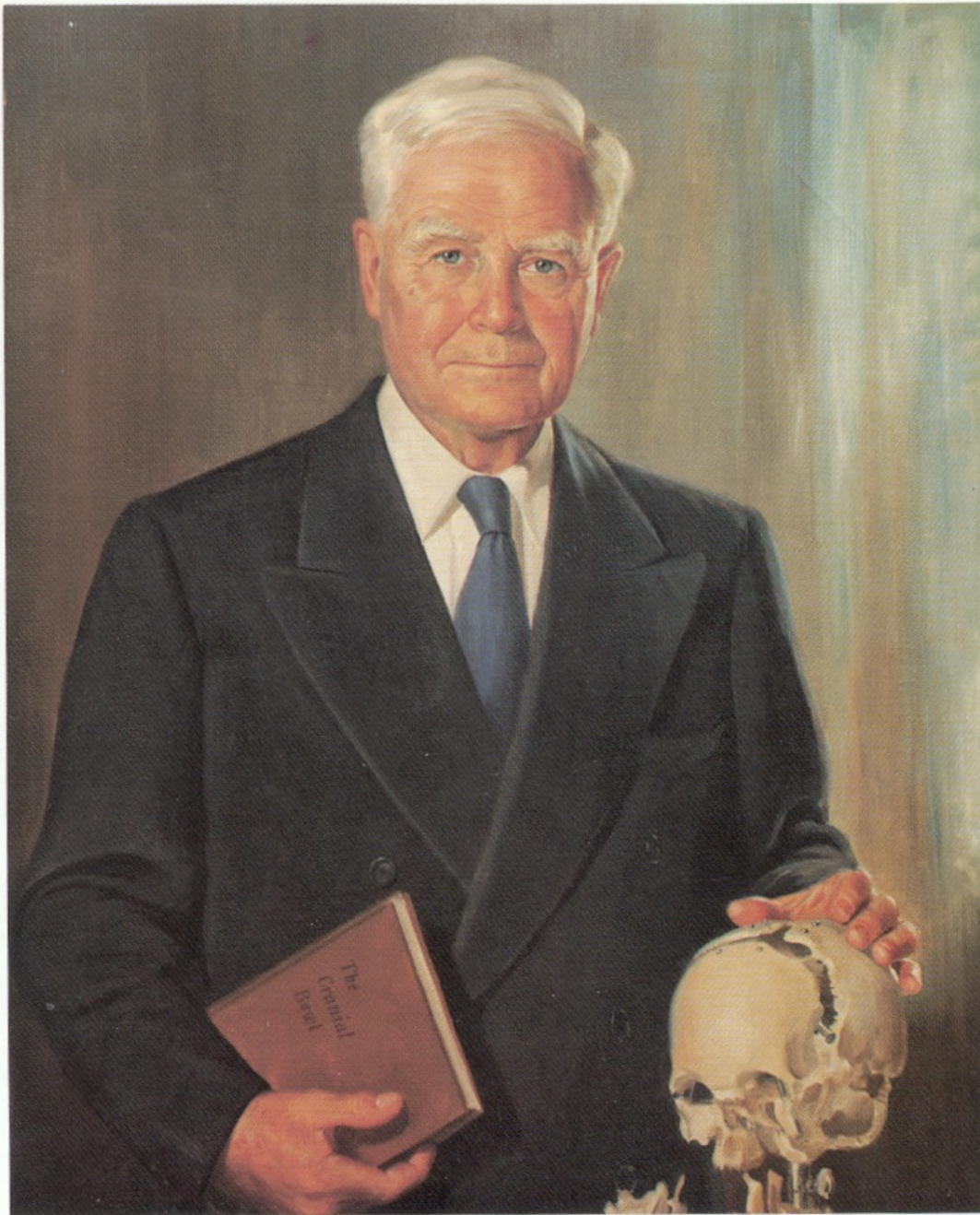
Natural Agencies
“Powers within the patient’s
body”



“Dr. Still has taken my hand in his and allowed me to feel the lesion as it was being exaggerated and *then as the natural agencies pulled the bones back into place.*”

COT p 112

- “The most any Physician can do for a patient is to render operative the forces within the body itself.”
- A.T.Still



“We do not force anything into place in the reduction of the lesion. We have something more potent than our own forces working always in the patient towards the direction of the normal.”

W. G. Sutherland

William Garner Sutherland, D.O., D.Sc.(hon.)

Natural Agencies

“Powers within the patient’s body”

- "Osteopathy is a science with possibilities as great as the magnitude of the heavens. It is a science dealing with the *natural forces* of the body. These *forces* within the patient are greater than any blind force that can safely be brought to bear from without."

WG Sutherland, DO

Natural Agencies

“Powers within the patient’s body”

- The Primary Respiratory Mechanism

Natural Agencies

“Powers within the patient’s body”

- “Respiratory co-operation is always with us in operations designed to utilize the powers within the patient’s body.”
- A.L. Wales, DO

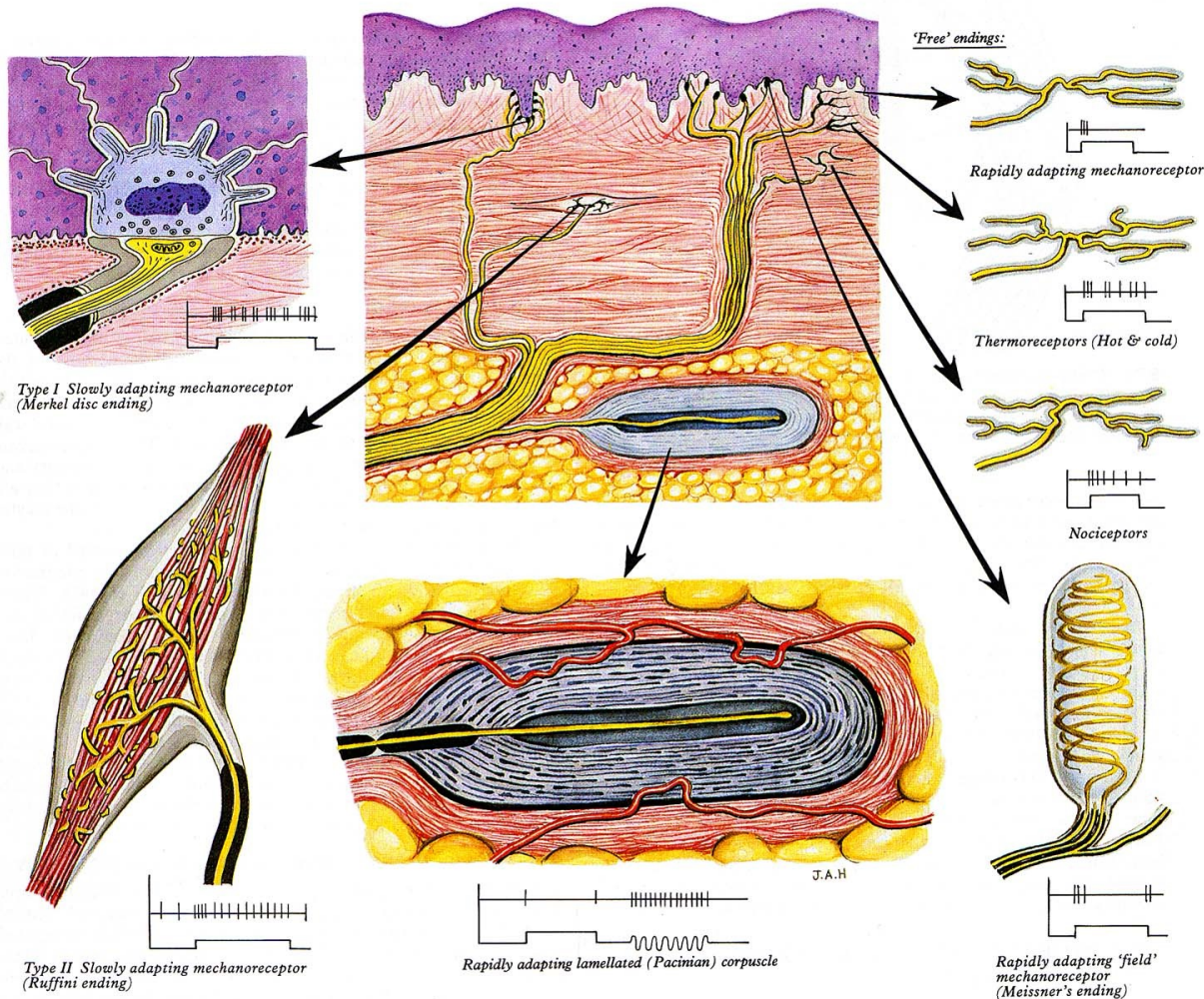
Balanced Tension

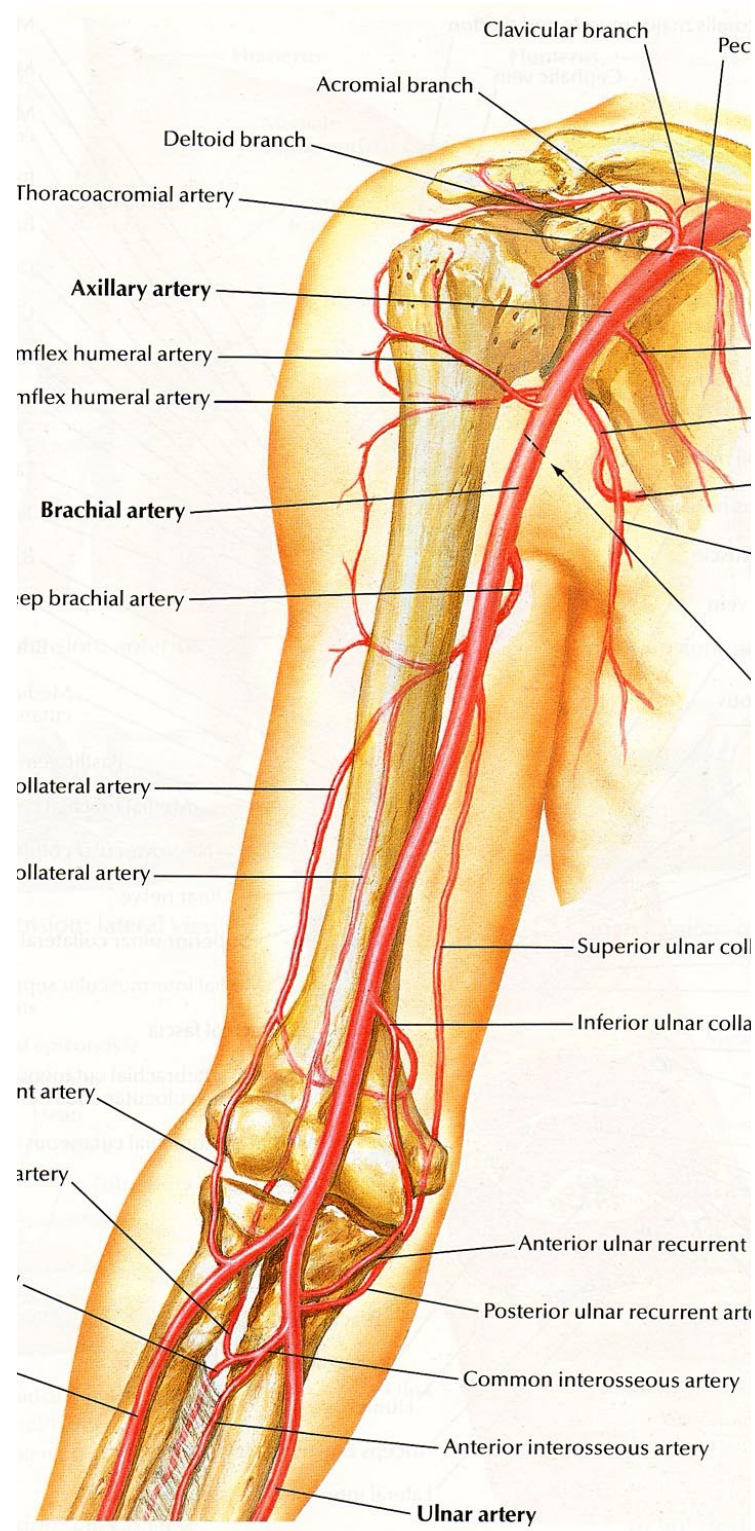
- There are a variety of inherent forces which can act directly on the areas of strain to correct the dysfunction.
- The action of these inherent forces is perceptible as the point of balanced tension is achieved.

The nervous system will respond to the point of balanced tension and “reset”, altered tissue tension to baseline and reducing facilitation

Proprioceptors

Nociceptors



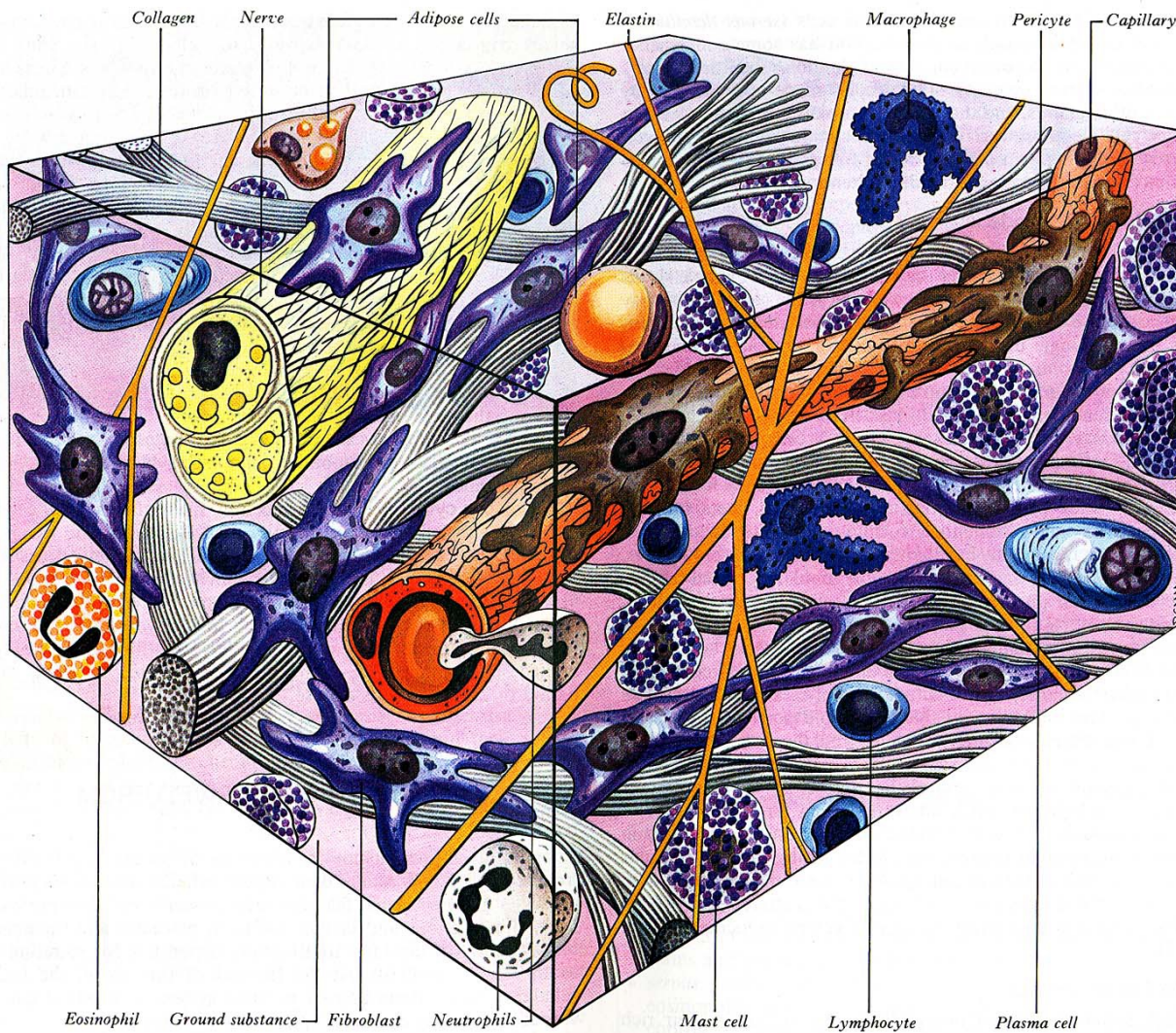


Local circulation,
venous and
lymphatic as well as
arterial will improve
as tissues are brought
to a point of
BLT/BMT



“We suffer from 2 causes,
the want of supply and the
burden of dead deposits”

A.T. Still, Autobiography



Interstitial fluid flow is also reduced by local tissue tension. This will have a direct effect on local cellular physiology and nutrition, an overlooked consequence of somatic dysfunction. The local strain will also reduce the expression of the intrinsic motions which produce the interstitial flux.

**The goal of treatment is
to improve the interchange of all the
fluids of the body
across all tissue interfaces.**

W.G. Sutherland

The evaluation of somatic dysfunction should include the impact on the local expression of inherent motion, such as respiration, pulse, and the primary respiratory mechanism.

This will provide evidence of the physiological impact the somatic dysfunction is having on local physiology.

BLT

Principles of Treatment



“It is my object in this work to teach principles as I understand them, and not rules. I do not instruct the student to punch or pull a certain bone, nerve or muscle for a certain disease, but by a knowledge of the normal and abnormal, give a specific knowledge for all diseases.”

A.T. Still, Philosophy of Osteopathy



BLT/BMT is the
treatment approach
taught by W.G.
Sutherland.



“Dr. Still has taken my hand in his and allowed me to feel *the lesion as it was being exaggerated* and then as the natural agencies pulled the bones back into place.”

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Principles of Treatment

- Dr. Sutherland used the principle of Balanced membranous and ligamentous Tension as his approach to treatment in the cranium and the body.
- His technique includes both direct and indirect methods
- The area of strain is brought to the point of balanced membranous/ligamentous tension
- All techniques involve engaging and utilizing the “powers within the patient's body”, from a “point of balanced tension”

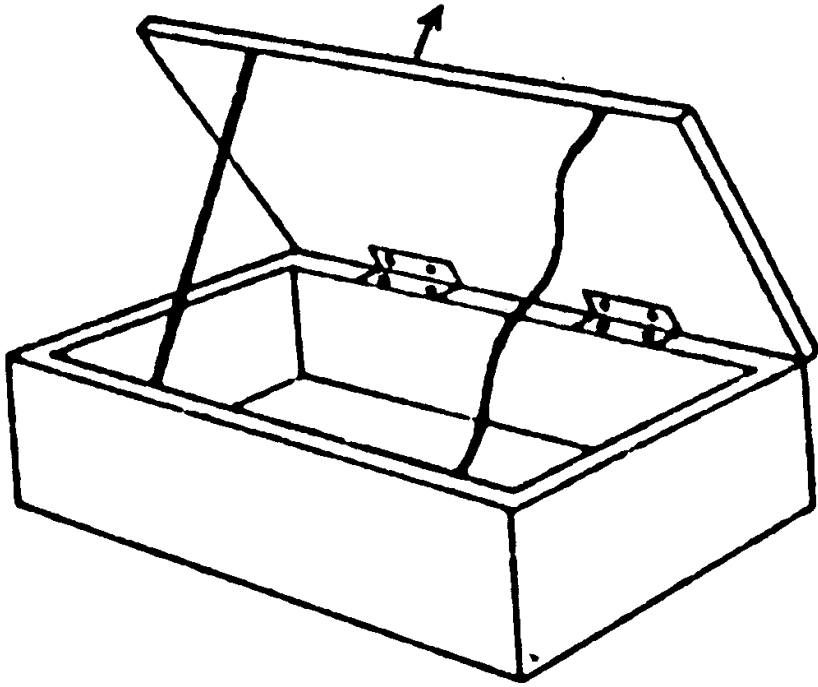
Balanced Ligamentous Tension

- The point of Balanced ligamentous/membranous tension is defined as, “the point in the range of motion of an articulation where the ligaments and membranes are poised between the normal tension present throughout the free range of motion and the increased tension preceding the strain...”

Balanced Ligamentous Tension

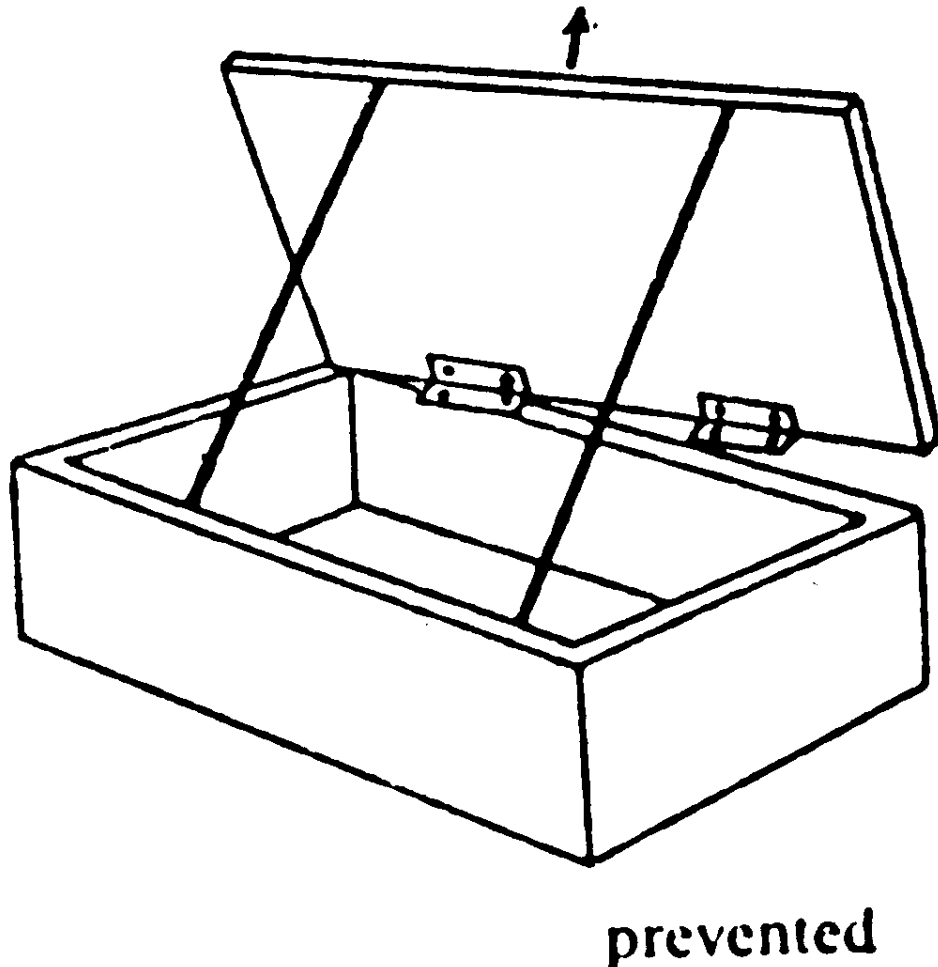
- Although the concept is simple, the depth lies in the precision with which it is applied, and the window into body physiology it affords
- The point of BLT/BMT may be achieved through both direct and indirect positioning

Balanced Ligamentous Tension



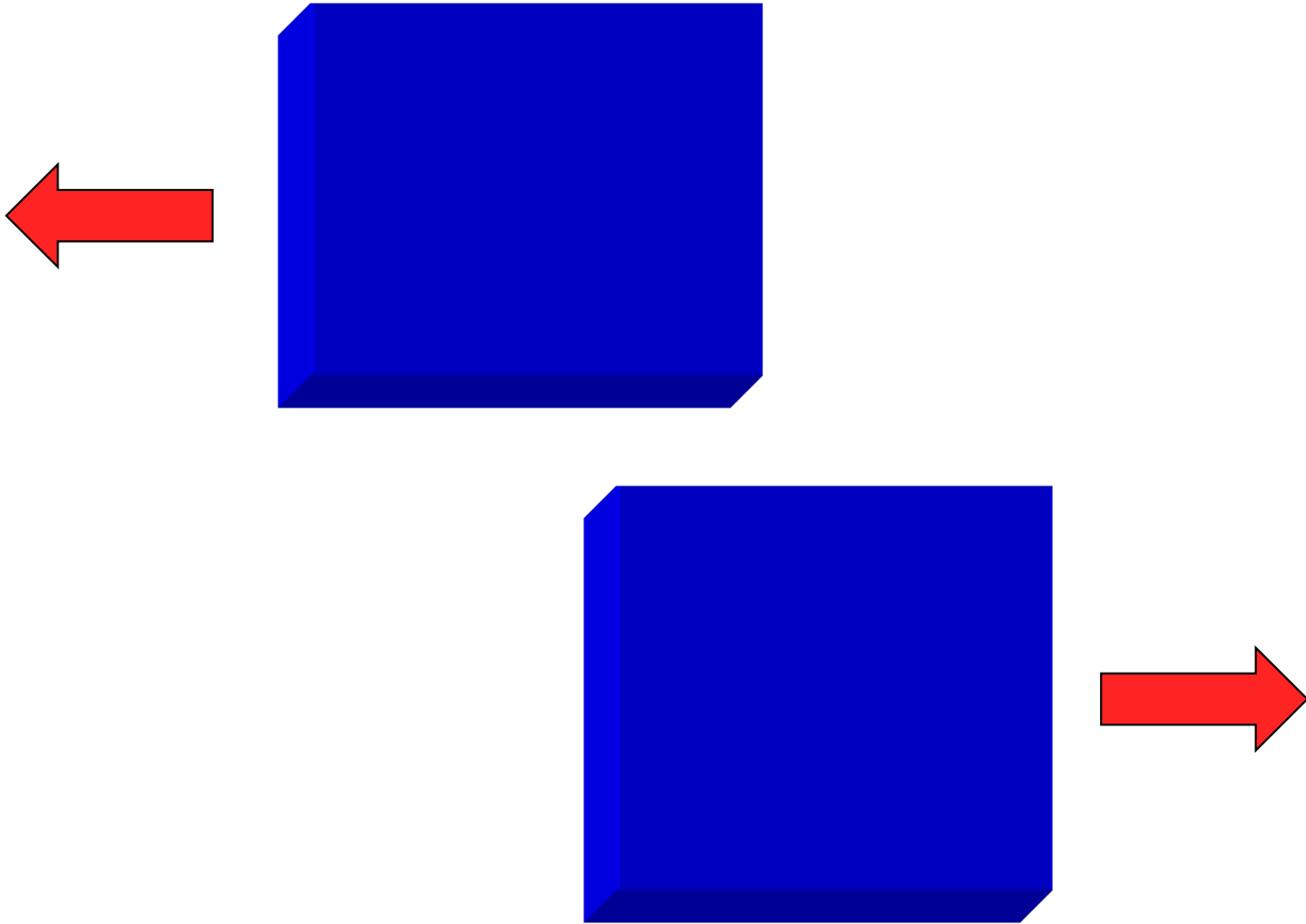
This might diagrammatically represent a ligamentous articular strain. By partially closing the lid and slacking both straps (ligaments), a point of balanced ligamentous tension would be achieved. This would represent an indirect approach to the principles of balanced ligamentous tension.

Balanced Ligamentous Tension

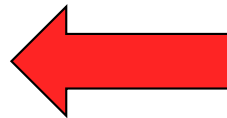
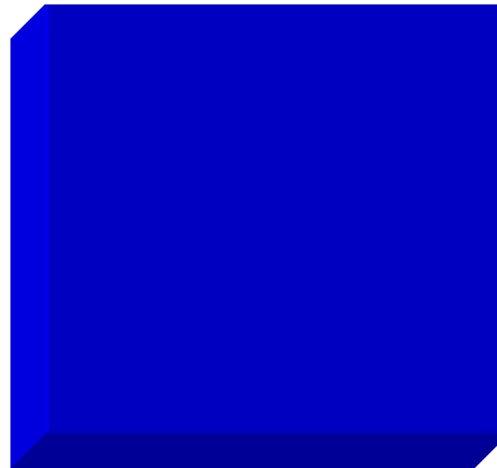
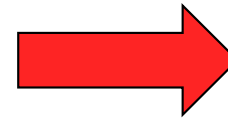


If the articulation is carried into the strain, the articulation would be displaced in such a way as to tense the slacked strap (ligament), and this would also bring this to a point of balanced tension. This point of balanced tension would be achieved through direct (into the barrier) positioning. When positioning in the direct direction, one does not attempt to overcome the resistance. Rather, one matches tension in a manner which allows the inherent force to make the correction.

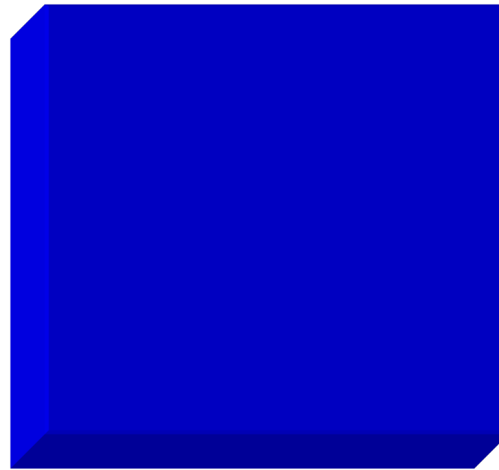
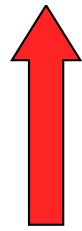
Exaggeration



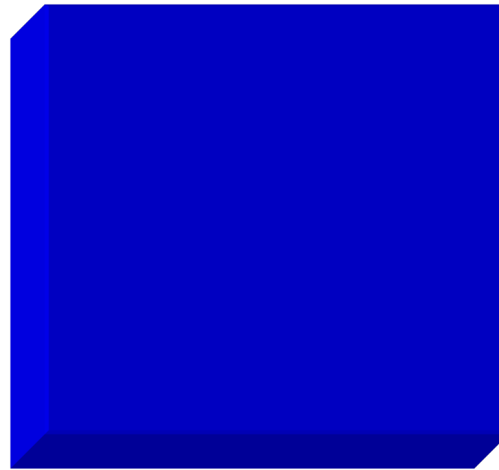
Direct Action



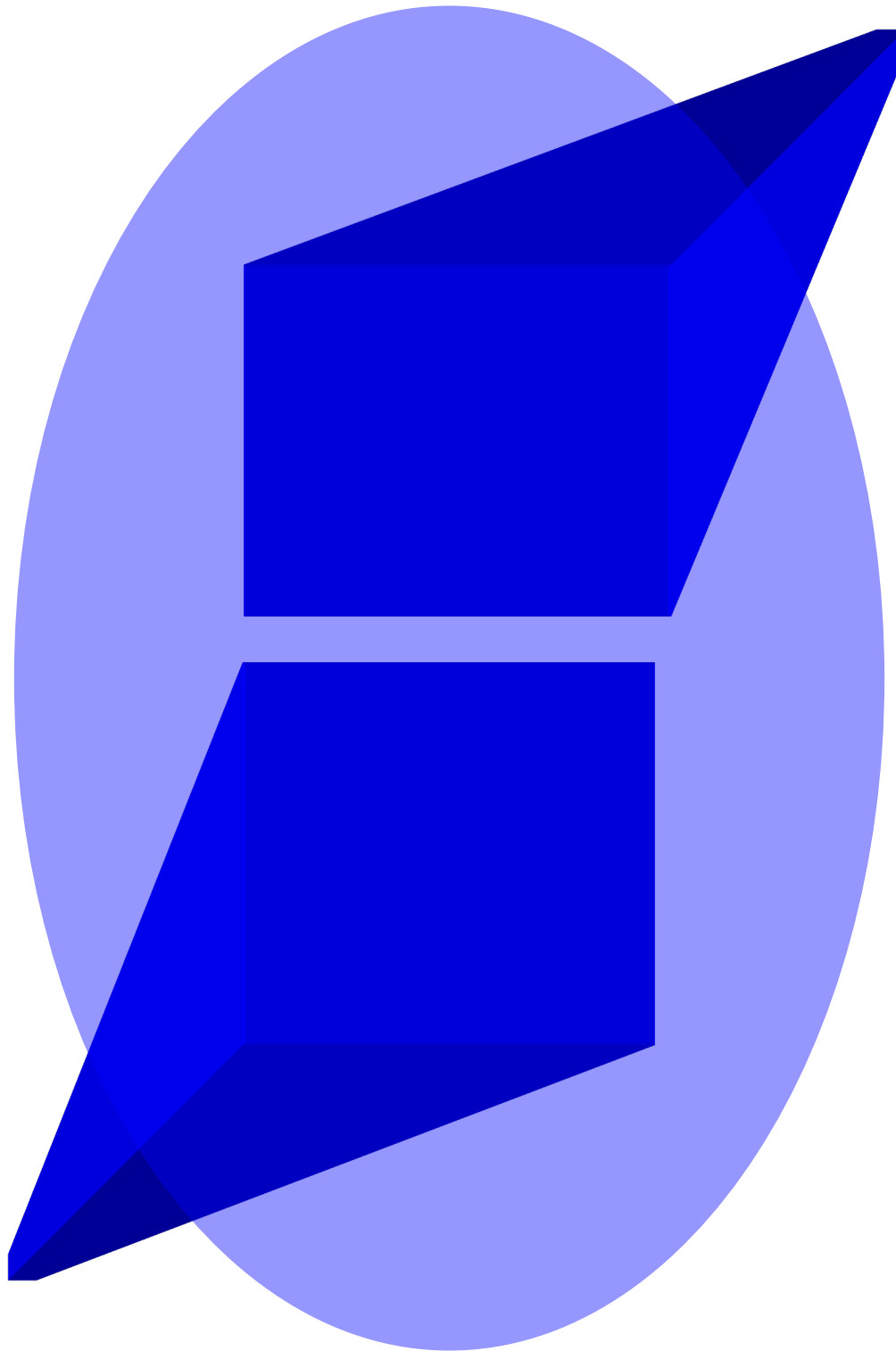
Disengagement



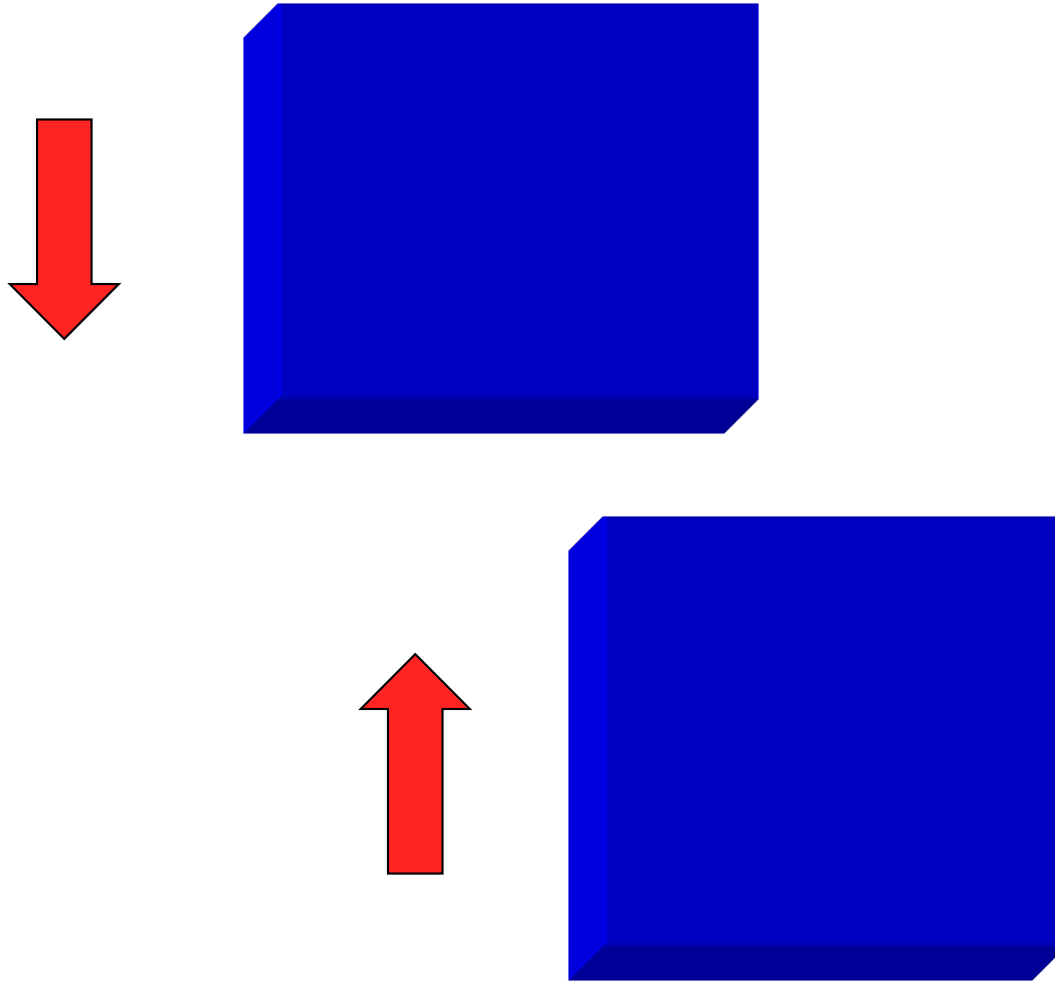
Opposing Physiologic Motion



Molding



Compression? - LAS



Assistance of Innate Forces

- **Directing the Tide**
- **Respiratory Assistance**
- **Postural Assistance**

Details of Treatment

- Diagnosis
- Planning
- Frequency
- Endpoints

Conclusions

- Balanced Membranous/ligamentous tension is a treatment principle which an area of strain is brought to a point of balanced tension to engage and utilize the inherent forces within the patients body to make the correction.

Conclusions

- Palpation throughout the treatment may offer a perception of the various physiologic processes occurring, underlying the correction, and a deeper understanding of body physiology.